Modern Business Cycle Theory

Measurement, expectations and the equilibrium real rate

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Why do economies fluctuate?

The economic question

Output, consumption and interest rates fluctuate. Which movements come from how we measure fluctuations, and which can be explained by choices and market clearing?

In this first session

Measure a cycle, derive a consumption–saving equilibrium, then obtain a computable decision rule. Distinguish the exact economic model from the approximation used to solve it.

$$\underbrace{y_t = \tau_t + \psi_t}_{\text{measurement}} \quad \longrightarrow \quad \underbrace{c_t = y_t,\ b_t = 0}_{\text{equilibrium}} \quad \longrightarrow \quad \underbrace{\hat{x}_t = P\hat{x}_{t-1} + Q\epsilon_t}_{\text{solution}}.$$

Plan

1. What is a business cycle?
2. Consumption–saving and general equilibrium
3. Approximation methods

First decide what the data call a cycle before assigning it an economic mechanism.

180 minutes of work, plus a 15-minute break

Measurement and filter interpretation	35 min
Household choices and market clearing	45 min
Expectations, risk and Exercise 1	30 min
Local approximation and Exercise 2	45 min
Dynare procedure and synthesis	25 min

The 180 minutes include 25 minutes of individual exercise work and oral checkpoints. Take a separate 15-minute break after the equilibrium block.

From dated recessions to frequency bands

Definition: business-cycle fluctuations

Coordinated expansions and contractions across many activities, not a clock with a fixed period. NBER and CEPR date recessions. The frequency convention here retains periods of 6–32 quarters.[a]

[a] Burns and Mitchell (1946). Handout §1.1.

Historical question	What changes in the analysis?
Mitchell, national accounts	Quantify aggregate activity and date expansions and recessions.
Juglar, Kuznets, Kondratiev	Ask whether investment or innovations generate regular periods.
Modern macroeconomics	Separate trend and cycle, then explain shocks and propagation.

A policy implication: the measured structural deficit depends on the chosen cyclical benchmark. Dating a recession and filtering GDP are related but different exercises.

From exponential growth to an additive cycle

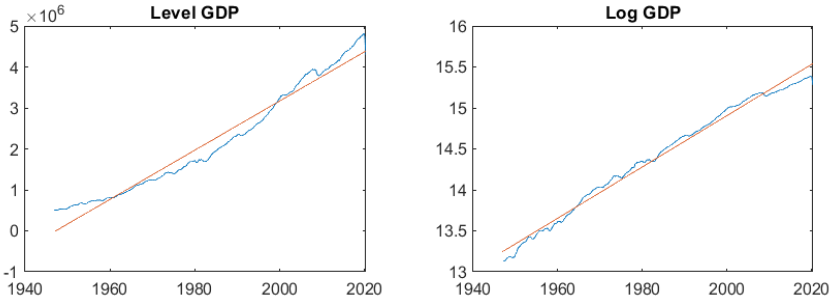
$$Y_t = Y_0(1+\gamma)^t \exp(\varepsilon_t)$$

- Y_t: real GDP in levels. γ: quarterly growth rate in decimal units.
- Take logs and define $y_t = \log Y_t$:

$$y_t = \log Y_0 + t\log(1+\gamma) + \varepsilon_t.$$

- For small γ, $\log(1+\gamma) \simeq \gamma$. The trend is approximately linear in time.
- A log deviation of 0.01 is approximately a one-percent deviation.

GDP in levels and logs



- A straight line in levels does not capture exponential growth.
- Logs help, but do not establish that the growth trend is constant.[1]

[1] Source: original handout, `logGDP` figure. Historical illustration, not regenerated data.

From fitted growth to the cyclical residual

The handout's decomposition is $y_t = \tau_t + \psi_t$, with $y_t = \log Y_t$.

$$\log Y_t = \log Y_0 + t \log(1+\gamma) + \varepsilon_t,$$
$$\widehat{\tau}_t = \widehat{\log Y_0} + t\widehat{\log(1+\gamma)},$$
$$\widehat{\psi}_t = y_t - \widehat{\tau}_t.$$

What OLS estimates

An intercept and one growth slope for the entire sample. For small growth, $\log(1+\gamma) \simeq \gamma$.

What a break does

After a permanent downward shift, the old extrapolated path can create a persistently negative measured cycle.

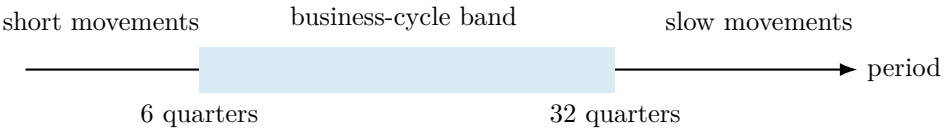
The HP filter: fit versus smoothness

$$\min_{\{\tau_t\}} \sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}) \right]^2.$$

- First term: keep the trend close to observed log GDP.
- Second term: penalize changes in trend growth. λ controls smoothness.
- The supplied quarterly example uses $\lambda = 1600$.
- This is a two-sided filter. Endpoints are fragile and estimates change when observations are added.[2]

[2] Hodrick and Prescott (1997). Critique discussed in Hamilton (2018).

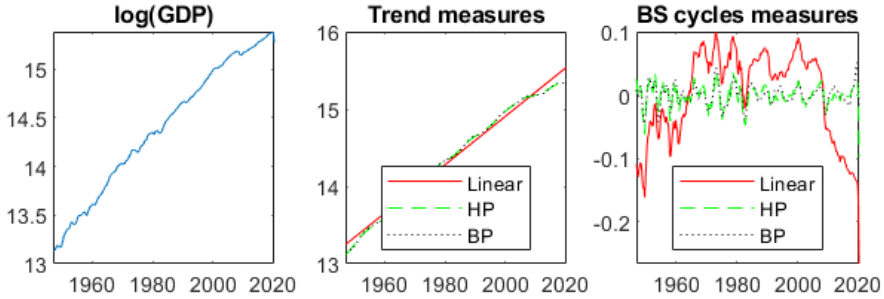
Select the business-cycle frequency band



- Retain the chosen band rather than fit a particular trend curve.
- `bpass(gdp,6,32)` returns the cycle, not the trend.
- Finite samples require approximations, especially near the boundaries.[3]

[3] Baxter and King (1999), Christiano and Fitzgerald (2003). Supplied implementation: Christiano–Fitzgerald approximation.

Three filters, three measured cycles



- Compare the trend lines before comparing the cyclical amplitudes.
- A long negative deviation from a fixed trend can coexist with a small HP cycle.
- Do not call one line the true cycle without stating the economic question.[4]

[4] Source: original handout, `BSmeasures` figure. Historical illustration, not regenerated data.

Read the three filter outputs

```
gdp = log(mydata(:,2));
cycle_lin = detrend(gdp,1);
trend_hp = hpfilter(gdp,'Smoothing',1600);
cycle_hp = gdp-trend_hp;
cycle_bp = bpass(gdp,6,32);
```

1. Run `logGDP.m` and `BSmeasures.m` from `session1/codes`.
2. Identify which objects are trends and which are cycles.
3. Compare standard deviations and endpoints on the same sample.[5]

[5] DBnomics series: OECD/QNA/USA.B1_GS1.LNBQRSA.Q. Internet and HP-filter toolbox required.

Compare filters in common units

- `detrend` returns the residual. Subtract it from log GDP to recover the linear trend.
- In the supplied HP call, the first output is the trend. Subtract it to obtain the cycle.
- `bpass` returns the band. Its complement includes slow and fast movements, not a pure low-frequency trend.
- Multiplying each log cycle by 100 makes the vertical axis approximately percent.
- Report the series, vintage, sample and filter. Do not infer a current-quarter result from the archived figure.

The regularities a model should explain

small innovations $\longrightarrow$ dynamic propagation $\longrightarrow$ joint fluctuations.

Regularity reported in the handout	Implication for a theory
Output co-moves across sectors	Explain common movements, not one isolated series.
Durables are more volatile	Explain different responses to a common impulse.
Prices and short rates are procyclical	Jointly determine quantities and prices.
Money and velocity are procyclical	Compare real and monetary co-movements.

Lucas's representative-agent, rational-expectations approach links these regularities to optimizing decisions and market clearing.[6]

[6] Frisch (1933), Blanchard and Watson (1986), Lucas (1977). These are the historical regularities discussed in handout §1.3, not restrictions imposed on every dataset.

Plan

1. What is a business cycle?
2. Consumption–saving and general equilibrium
3. Approximation methods

A measured pattern is not yet a mechanism. Derive household choices and market clearing.

The endowment economy: quantities and prices

- Fruit falls from trees. The household cannot choose aggregate production.
- Financial claims let a household contemplate shifting consumption across dates.
- But all households are identical, and the economy has no outside lender.
- The key unknown is the interest rate that makes zero net saving optimal.

Do not add capital or a production choice to this model. Those enter in Session 2.

Persistence in the endowment process

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2).$$

- y_t: fruit endowment. Its deterministic steady state is one.
- $0 \leq \rho < 1$: persistence. σ: innovation standard deviation.
- A weather shock changes current resources and forecasts of future resources.
- Additive Gaussian shocks are a local approximation. They permit non-positive consumption and do not define a finite inverse-consumption expectation.

Preferences across dates and states

$$\max_{\{c_{t+s},b_{t+s}\}_{s\geq 0}} E_t \sum_{s=0}^{\infty} \beta^s \log c_{t+s}, \qquad 0 < \beta < 1.$$

- c_t: consumption. b_t: bonds chosen today and held to tomorrow.
- β: discount factor. Lower β means greater impatience.
- Concavity makes an additional unit more valuable when consumption is low.
- E_t: expectation using information available at date t, not knowledge of future shocks.

The bond return and the budget constraint

$$c_t + b_t = r_{t-1} b_{t-1} + y_t.$$

- At the start of t, yesterday's bond pays $r_{t-1} b_{t-1}$.
- At t, choose consumption and new bond holdings after observing y_t.
- r_t is the gross risk-free real return, known at t and paid at $t+1$.
- The net real interest rate is $r_t - 1$, not r_t.

The constraint binds because utility is increasing in consumption.

The household Lagrangian

$$\mathcal{L}_t = E_t \sum_{s=0}^{\infty} \beta^s \left[\log c_{t+s} - \lambda_{t+s} \left(c_{t+s} + b_{t+s} - r_{t+s-1} b_{t+s-1} - y_{t+s} \right) \right].$$

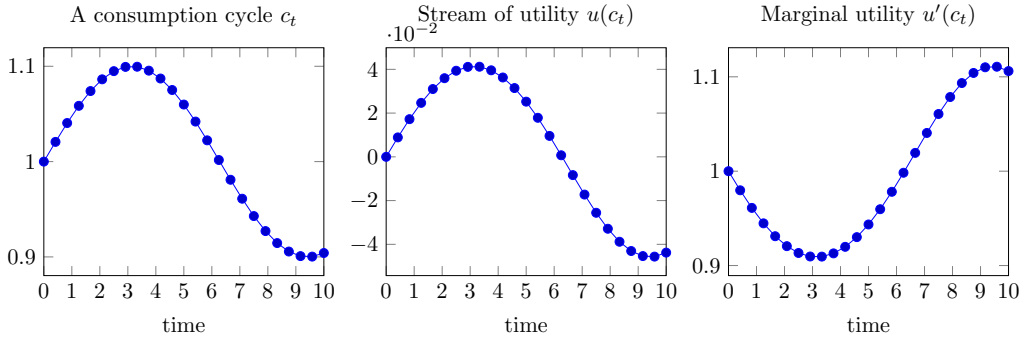
- λ_t: marginal value of relaxing the period-t budget.
- Discounting the constraint is a normalization of the multiplier.
- Differentiate with respect to consumption and the bond choice.
- Interior bond choice, a no-Ponzi restriction and transversality are part of the optimization problem.

Consumption: marginal benefit equals cost

$$\frac{\partial \mathcal{L}_t}{\partial c_t} = 0 \qquad \Longrightarrow \qquad \lambda_t = u'(c_t) = \frac{1}{c_t}.$$

- One more unit of consumption yields $1/c_t$ extra utility.
- One more unit uses resources worth λ_t.
- At the optimum, marginal benefit equals marginal resource cost.
- Higher consumption means lower marginal utility, even though total utility rises.

Consumption, utility and marginal utility



- Read the consumption peak, then locate the marginal-utility trough.
- This reversal drives the household's willingness to save.[7]

[7] Source: original handout, illustrative consumption cycle under log utility. Historical illustration, not an empirical estimate.

The bond choice links two dates

$$-\lambda_t + \beta r_t E_t \lambda_{t+1} = 0.$$

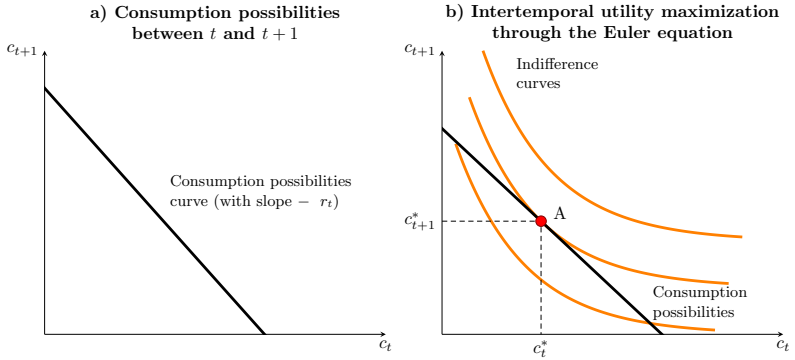
- Buying one more bond costs one unit of resources today: loss λ_t.
- It delivers r_t units tomorrow: expected gain $\beta r_t E_t \lambda_{t+1}$.
- r_t is known today, so it can be taken outside the expectation.
- An interior optimum equates these two marginal values.

Euler and the stochastic discount factor

$$\frac{1}{c_t} = \beta r_t E_t \left[\frac{1}{c_{t+1}}\right], \qquad 1 = r_t E_t m_{t+1}, \qquad m_{t+1} = \beta \frac{c_t}{c_{t+1}}.$$

- m_{t+1}: stochastic discount factor, the value of future consumption relative to current consumption.
- A high expected discount factor implies a low equilibrium gross rate.
- In a deterministic two-date comparison, $r_t = u'(c_t)/[\beta u'(c_{t+1})]$.

Intertemporal opportunities and preferences



- Saving one unit today delivers r_t units tomorrow. The budget slope is $-r_t$.
- The tangency expresses the deterministic Euler condition.[8]

[8] Source: original handout, two-period diagram. Historical illustration, not a complete picture of stochastic equilibrium.

Tangency and the Euler condition

For the deterministic two-date diagram, consider a small feasible reallocation.

$$\begin{aligned} \text{Budget slope:} \quad & dc_{t+1} = -r_t\, dc_t, \\ \text{Same utility:} \quad & u'(c_t)\, dc_t + \beta u'(c_{t+1})\, dc_{t+1} = 0, \\ \text{Tangency:} \quad & r_t = \frac{u'(c_t)}{\beta u'(c_{t+1})}. \end{aligned}$$

From the diagram to uncertainty

Tomorrow's marginal utility is random, while the bond return is fixed today:

$$u'(c_t) = \beta r_t E_t[u'(c_{t+1})].$$

The relevant forecast is marginal utility, not just consumption.

Market clearing selects the interest rate

$$b_t = 0 \qquad \Longrightarrow \qquad c_t = y_t.$$

- No external borrowing or lending. Domestic bonds have zero net supply.
- With identical households, each household also holds zero bonds in equilibrium.
- Aggregate consumption must equal the available fruit.
- The rate adjusts so that households choose not to trade at that allocation.

Two equations close the equilibrium

Substitute goods-market clearing $c_t = y_t$ into the Euler condition:

$$\frac{1}{y_t} = \beta r_t E_t \left[\frac{1}{y_{t+1}} \right].$$

The formal Euler benchmark

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t, \\ r_t = \left[\beta y_t E_t \{ y_{t+1}^{-1} \} \right]^{-1}. \end{cases}$$

The first equation fixes the allocation. The second selects the rate at which desired saving is zero. More uncertainty affects the rate through expected marginal utility. Positive consumption and a well-defined expectation are required.

Why expectations change decisions today

The handout gives two mechanisms absent from a purely backward-looking model.[9]

Price and wage setting

Monetary accommodation today $\longrightarrow$ higher expected future inflation $\longrightarrow$ a wage demand today that compensates for the expected loss of purchasing power.

International portfolio allocation

A higher domestic interest rate $\longrightarrow$ immediate currency appreciation and expected subsequent depreciation, holding the foreign rate and expected future exchange rate fixed.

The expected exchange-rate loss offsets the higher domestic interest return. In both examples, expectations enter a current choice or market price.

[9] Handout §2.4. Lucas (1976), Muth (1961).

Rational expectations and forecast errors

$$
\begin{aligned}
E_t y_{t+1} &= E_t[1 - \rho + \rho y_t + \varepsilon_{t+1}] \\
&= 1 - \rho + \rho y_t, \\
y_{t+1} - E_t y_{t+1} &= \varepsilon_{t+1}.
\end{aligned}
$$

- The household knows the process and current endowment, not tomorrow's innovation.
- Forecast errors have conditional mean zero. Individual errors need not shrink over time.
- Expectations change immediately when today's endowment changes.[10]

[10] Muth (1961), Lucas (1976).

Exact expectations versus certainty equivalence

$$E_t\left[\frac{1}{y_{t+1}}\right] \neq \frac{1}{E_t y_{t+1}} \quad \text{in general.}$$

- Replace future output by its conditional mean to obtain the teaching rule

$$r_t^{\mathrm{CE}} = \frac{1-\rho+\rho y_t}{\beta y_t}.$$

- This is correct to first order around the deterministic steady state, not an exact treatment of risk.
- With positive consumption and finite expectations, convexity gives $E_t[1/y_{t+1}] \geq 1/E_t y_{t+1}$.

Why more fruit lowers the real rate

$$r_t^{\mathrm{CE}} = \frac{\rho}{\beta} + \frac{1-\rho}{\beta y_t}, \qquad \frac{\partial r_t^{\mathrm{CE}}}{\partial y_t} = -\frac{1-\rho}{\beta y_t^2} < 0.$$

- More fruit today raises desired saving because part of the increase is temporary.
- Aggregate saving cannot rise. The rate must fall to discourage it.
- If ρ is close to one, expected future fruit rises almost as much as current fruit. The rate response is weaker.

Read and work: the nonlinear CE model

Open Handout 1, Exercise 1

Read §2.1–2.4, then work questions 1–2 of *Simulating the certainty-equivalent model*. Allow 8 minutes to design the recursion and plot.

- Use the exercise's 400-period calibration, not the script's 10,000-period horizon.
- Check steady state, innovation indexing and positive endowments before interpreting rates.
- Keep r_t as a gross-rate level, not the deviation introduced in §3.

Read and work: moments and nonlinear responses

Continue Handout 1, Exercise 1

Read questions 3–7. Spend 7 minutes on a common-units comparison and the signs of the positive and negative IRFs.

- Compare a log endowment deviation with a log-GDP cycle, not levels with percentages.
- Separate innovation volatility from propagation persistence.
- Inspect nonlinear curvature and histograms only on admissible positive paths.

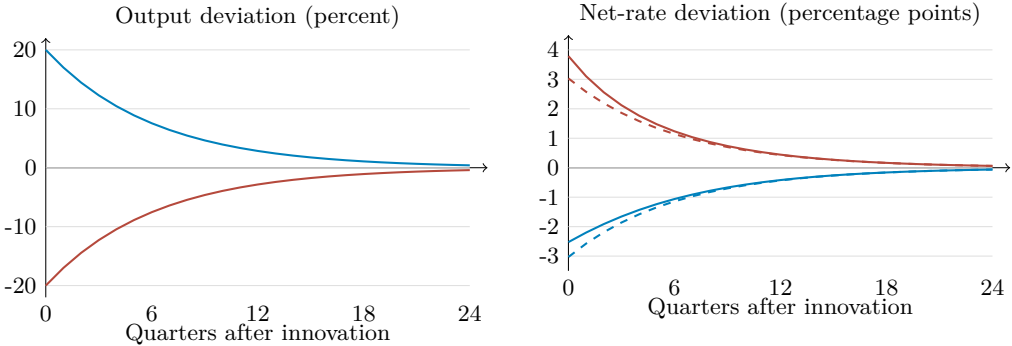
Separate persistence from volatility

```
for t = 2:T+1
    y(t) = 1-rho+rho*y(t-1)+e(t-1);
end
r = (1-rho+rho*y)./(beta*y);
```

$$\mathrm{Var}(y_t) = \frac{\sigma^2}{1-\rho^2}, \qquad \mathrm{Corr}(y_t, y_{t-1}) = \rho.$$

- For the exercise calibration, the stationary output standard deviation is about 0.190 in level units.
- Match the filter and units before comparing to a GDP cycle. These illustrative parameters are not an empirical estimate.

Nonlinear and linear impulse responses



- Blue: +0.20 endowment innovation. Red: −0.20. No later shocks.
- Solid: certainty-equivalent rule. Dashed: first-order rate approximation.[11]

[11] Analytical illustration generated from the displayed equations. $\beta = .99$, $\rho = .85$. Rates are deviations in percentage points, not empirical IRFs.

Risk enters through marginal utility

Definition: CRRA curvature

The handout writes $u'(c) = c^{-\sigma_C}$, with $\sigma_C > 0$. Curvature σ_C is not the innovation standard deviation σ.

$$y_t^{-\sigma_C} = \beta r_t E_t[y_{t+1}^{-\sigma_C}],$$
$$r_t = \frac{y_t^{-\sigma_C}}{\beta E_t[y_{t+1}^{-\sigma_C}]}.$$

Replacing future output by its conditional mean gives

$$r_t^{\mathrm{CE}} = \frac{1}{\beta}\left(\frac{E_t y_{t+1}}{y_t}\right)^{\sigma_C}.$$

For $\sigma_C = 1$ this recovers the log-utility teaching rule. Certainty equivalence captures first-order movements, not generally risk corrections.

Plan

1. What is a business cycle?
2. Consumption–saving and general equilibrium
3. Approximation methods

Equilibrium is defined. We now need approximations that preserve its economic restrictions.

Equilibrium restrictions and decision rules

$$E_t f_\theta(x_{t+1}, x_t, x_{t-1}, \epsilon_t) = 0, \qquad x_t = g(x_{t-1}, \epsilon_t).$$

- x_t: model variables. θ: structural parameters. f_θ: equilibrium restrictions.
- g: rule that satisfies those restrictions along every admissible history.
- Larger models rarely have an explicit nonlinear rule.
- Approximate around a known reference point, then solve the resulting linear system.

Substitute the rule into equilibrium

The starting objects are equations (16)–(17):

$$E_t[f_\theta(x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0, \qquad x_t = g(x_{t-1}, \epsilon_t).$$

One substitution, equation (18), keeps tomorrow's innovation random:

$$E_t[f_\theta(g(x_t, \epsilon_{t+1}), x_t, x_{t-1}, \epsilon_t)] = 0.$$

Substitute again to obtain equation (19):

$$0 = E_t\big[f_\theta\big(g(g(x_{t-1}, \epsilon_t), \epsilon_{t+1}), \\ g(x_{t-1}, \epsilon_t), x_{t-1}, \epsilon_t\big)\big].$$

The unknown is a function g that satisfies this restriction for every admissible state and innovation. Setting $\epsilon_{t+1} = 0$ inside a nonlinear expression would discard risk.

The toy model gives an explicit policy rule

For the certainty-equivalent version, substitute current endowment into the rate rule:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t,$$
$$r_t^{\mathrm{CE}} = \frac{\rho}{\beta} + \frac{\beta^{-1}(1-\rho)}{1 - \rho + \rho y_{t-1} + \varepsilon_t}.$$

Reading equations (20)–(21)

Both current variables are now functions of yesterday's state y_{t-1} and today's innovation. Yesterday's rate is not an independent state.

In larger models, solving the functional restriction at every possible state is not tractable. We approximate locally, then solve the resulting linear rational-expectations system.

Taylor expansion and log deviations

$$f(x_t) - f(\bar{x}) \simeq f'(\bar{x})(x_t - \bar{x}).$$

- Evaluate the slope at the reference point $\bar{x}$.
- Define a log deviation: $\hat{x}_t = \log(x_t/\bar{x})$.
- For a small deviation, $x_t - \bar{x} \simeq \bar{x}\hat{x}_t$, giving

$$f(x_t) - f(\bar{x}) \simeq f'(\bar{x})\bar{x}\hat{x}_t.$$

- $100\hat{x}_t$ is an approximate percentage deviation, not a level.

Two variables: keep both marginal effects

Start from the ordinary first-order expansion about $(\bar{x}, \bar{y})$:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})(x_t - \bar{x}) + f'_y(\bar{x}, \bar{y})(y_t - \bar{y}).$$

Use $x_t - \bar{x} \simeq \bar{x}\hat{x}_t$ and $y_t - \bar{y} \simeq \bar{y}\hat{y}_t$:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})\bar{x}\hat{x}_t + f'_y(\bar{x}, \bar{y})\bar{y}\hat{y}_t.$$

Example: a product

For $f(x, y) = xy$, $f'_x = \bar{y}$ and $f'_y = \bar{x}$. Therefore

$$x_t y_t - \bar{x}\bar{y} \simeq \bar{x}\bar{y}(\hat{x}_t + \hat{y}_t).$$

The handout's six linearization rules

Each right-hand side approximates the expression minus its steady-state value.

Level expression	First-order deviation
x_t	$\bar{x}\hat{x}_t$
$x_t y_t$	$\bar{x}\bar{y}(\hat{x}_t + \hat{y}_t)$
x_t / y_t	$(\bar{x}/\bar{y})(\hat{x}_t - \hat{y}_t)$
x_t^a	$\bar{x}^a a \hat{x}_t$
$\log(x_t)$	$\hat{x}_t$
$\exp(x_t)$	$\exp(\bar{x})\bar{x}\hat{x}_t$

A sum retains the steady-state weights of its terms. A zero-mean additive innovation stays in levels, since its log is not defined.

Application: the deterministic steady state

$$\bar{y} = 1 - \rho + \rho\bar{y} \qquad \Longrightarrow \qquad \bar{y} = 1,$$
$$\bar{r} = \frac{1 - \rho + \rho\bar{y}}{\beta\bar{y}} \qquad \Longrightarrow \qquad \bar{r} = \beta^{-1}.$$

- Set innovations to zero and variables constant across dates.
- With $\beta = .99$, the gross quarterly rate is approximately 1.0101.
- The net quarterly rate is approximately 1.01%.
- The deterministic steady state is the expansion point, not necessarily the stochastic mean of a nonlinear variable.

Application: linearize endowment

$$y_t - 1 = \rho(y_{t-1} - 1) + \varepsilon_t,$$
$$\hat{y}_t \simeq \rho \hat{y}_{t-1} + \varepsilon_t.$$

- $\bar{y} = 1$ converts the level deviation into a log deviation to first order.
- The constant $1 - \rho$ disappears after subtracting the steady state.
- The innovation remains additive and is not log-transformed.
- The approximate response to one innovation is geometric decay at rate ρ.

Application: linearize the gross rate

$$
\begin{aligned}
r_t^{\mathrm{CE}} &= \frac{\rho}{\beta} + \frac{1-\rho}{\beta y_t}, \\
\bar{r}\hat{r}_t &\simeq -\frac{1-\rho}{\beta \bar{y}}\hat{y}_t, \\
\hat{r}_t &\simeq -(1-\rho)\hat{y}_t.
\end{aligned}
$$

- Use $\bar{y} = 1$ and $\bar{r} = 1/\beta$ in the last step.
- A one-percent output deviation lowers the *gross rate* by approximately $(1-\rho)$ percent relative to its steady state.
- For percentage-point changes in the net rate, use $100\bar{r}\hat{r}_t$.

The first-order stochastic Euler equation

$$-\hat{y}_t = \hat{r}_t - E_t\hat{y}_{t+1} \qquad \Longrightarrow \qquad \hat{r}_t = E_t\hat{y}_{t+1} - \hat{y}_t.$$

- Log utility makes marginal utility move approximately one-for-one against consumption.
- Rational expectations gives $E_t\hat{y}_{t+1} = \rho\hat{y}_t$.
- Hence $\hat{r}_t = -(1-\rho)\hat{y}_t$.
- Risk corrections disappear at first order, not from the underlying nonlinear economic problem.

From simultaneous equations to a decision rule

$$\underbrace{\begin{bmatrix}\hat{y}_t \\ \hat{r}_t\end{bmatrix}}_{\hat{x}_t} = \underbrace{\begin{bmatrix}\rho & 0 \\ -(1-\rho)\rho & 0\end{bmatrix}}_{P} \underbrace{\begin{bmatrix}\hat{y}_{t-1} \\ \hat{r}_{t-1}\end{bmatrix}}_{\hat{x}_{t-1}} + \underbrace{\begin{bmatrix}1 \\ -(1-\rho)\end{bmatrix}}_{Q} \varepsilon_t.$$

- Substitute the output recursion into the contemporaneous rate equation.
- The lagged rate has no independent role. The second column of P is zero.
- Eigenvalues are ρ and zero. For $|\rho| < 1$, responses decay.

An impulse response isolates propagation

$$\hat{x}_0 = 0, \quad \varepsilon_1 = d, \quad \varepsilon_t = 0 \ (t > 1) \qquad \Longrightarrow \qquad \hat{x}_{1+h} = P^h Q d.$$

- d: chosen shock size. h: quarters after the innovation.
- The first response is Qd. Subsequent responses multiply by P.
- At first order, doubling d doubles every response.
- Changing the sign of d reverses every response. Linear IRFs are symmetric.

Read and work: the linear approximation

Open Handout 1, Exercise 2

Read §3.2–3.3 and all six questions of *Simulating the linear model*. Use 10 minutes to implement the rule and compare with Exercise 1.

- Reuse the same innovations. Compare both rates in levels.
- Check symmetric linear IRFs and Gaussian linear distributions.
- Reduce shock size and explain how the mean squared approximation error changes.

Implementation: compare rates in levels

```
P = [rho 0; -(1-rho)*rho 0];
Q = [1; -(1-rho)];
x = zeros(2,T+1);
for t = 2:T+1
    x(:,t) = P*x(:,t-1)+Q*e(t-1);
end
r_linear = (1/beta)*(1+x(2,:)');
```

1. Reuse the same innovations as in the nonlinear CE simulation.
2. Compare rates in levels, then compare positive and negative IRFs.
3. Compute `mean((r-r_linear).^2)`. Reduce the shock scale and repeat.

Curvature and approximation error

$$r_t^{\mathrm{CE}} = \frac{\rho}{\beta} + \frac{1-\rho}{\beta y_t}, \qquad \frac{\partial^2 r_t^{\mathrm{CE}}}{\partial y_t^2} = \frac{2(1-\rho)}{\beta y_t^3} > 0.$$

- For positive output, a contraction raises the rate more than an equally sized expansion lowers it.
- The tangent rule omits that curvature and yields symmetric responses.
- Away from the zero-consumption boundary, smaller deviations reduce the local approximation error.
- Never compare a gross-rate level directly with its log deviation.

Dynare: the positive-endowment variant

```
model;
    y = y(-1)^rho*exp(e);
    1 = beta*r*(y/y(+1))^sigma;
end;
```

- This is a positive-endowment variant, not the additive process in the handout.
- `sigma`: CRRA curvature. Set it to one for log utility.
- In stochastic solution, the lead term enters the expected equilibrium residual.
- The file uses this Euler residual. At first order, greater current endowment lowers the gross rate.

A positive variant has an exact risk correction

For $y_{t+1} = y_t^{\rho} \exp(\varepsilon_{t+1})$ and Gaussian innovations,

$$E_t[y_{t+1}^{-\sigma_C}] = y_t^{-\rho\sigma_C} \exp(\sigma_C^2 \sigma^2/2),$$
$$r_t = \beta^{-1} y_t^{\sigma_C(\rho-1)} \exp(-\sigma_C^2 \sigma^2/2).$$

- Use the normal exponential moment, $E[e^{a\varepsilon}] = e^{a^2\sigma^2/2}$.
- This checks the lognormal variant, not the additive Gaussian model.
- Higher shock variance lowers the risk-free rate at fixed current endowment. The correction is second order in shock size.

Run and check the Dynare variant

1. Add the installed Dynare MATLAB folder to the path. Set the current folder to `session1/codes`.
2. Open `dyn_toy_model.mod`. Identify `sigy` (shock SD) and `sigma` (curvature).
3. In a working copy, use curvature one for the log benchmark. Run:

```
steady;
resid;
check;
stoch_simul(order=1,irf=30) y r;
```

Run `dynare dyn_toy_model`. Compare named IRFs and units. Then use order two for risk, keeping the endowment process unchanged.

Do not confuse the supplied model variants

- `toy_model.m`: additive endowment, CE rate, 10,000 simulated periods. The handout exercise asks for 400.
- `toy_model_bootstrap.m`: uses conditional Monte Carlo draws, not a residual bootstrap. Finite-sample positivity does not repair the Gaussian expectation.
- `dyn_toy_model.mod`: positive endowment, curvature 10 and order-two simulation. It is not a silent substitute for the log-utility benchmark.
- Report the equation, information set and approximation order before interpreting any difference as a risk effect.

Read the handout after class

- Read Handout 1 §1.1–1.3 for historical motivation and filtering caveats.
- Reconstruct Euler and equilibrium in §2.1–2.5 without the slides.
- Complete Exercise 1, questions 1–7, and Exercise 2, questions 1–6. Bring the common-shock rate comparison and your interpretation.
- Use §3.1–3.3 to distinguish economic equilibrium, local approximation and recursive solution.

Measurement, equilibrium, approximation

1. A measured cycle depends on a frequency and trend convention.
2. In the endowment economy, $c_t = y_t$ and $b_t = 0$. The rate clears the bond market.
3. Rational expectations uses the conditional distribution, not perfect foresight.
4. Certainty equivalence and first-order approximation do not capture every risk effect.
5. The decision rule $\hat{x}_t = P\hat{x}_{t-1} + Q\varepsilon_t$ is the bridge to simulation and, later, likelihood evaluation.

Next: make resources endogenous by allowing investment in physical capital.